

## **Household Behavior over the Life Cycle**

### Assignment 2: Household Labor Supply and Taxes

## Questions

1. The household utility function in Borella, De Nardi and Yang (2023) and in our toy model from lecture 6 is the sum of individual utilities. Change the household utility function to be a **weighted sum** of individual utilities. Let  $\mu$  be the weight on member 1 and  $1 - \mu$  be the weight on member 2. Write out the affected equations and implement it in Python. Show that the simulated labor market behavior is the same as in the baseline (from the lecture) when  $\mu = 0.5$ .

[Fix  $\mu = 0.5$  in the following.]

We want to make the preferences a weighted sum of the individual utilities with weights  $\mu$  and  $1 - \mu$ , which can be expressed as:

$$U(c_t, h_{1,t}, h_{2,t}) = \mu \cdot u_1(c_t, h_{1,t}) + (1 - \mu) \cdot u_2(c_t, h_{2,t})$$

Where the utility function for each member is:

$$u_j(c_t, h_{j,t}) = \frac{(c_t/2)^{1+\eta}}{1+\eta} - \rho_j \frac{h_{j,t}^{1+\gamma}}{1+\gamma}$$

Inserting this into the weighted sum of the individual utilities:

$$U(c_t, h_{1,t}, h_{2,t}) = \mu \cdot \left[ \frac{(c_t/2)^{1+\eta}}{1+\eta} - \rho_1 \frac{h_{1,t}^{1+\gamma}}{1+\gamma} \right] + (1 - \mu) \cdot \left[ \frac{(c_t/2)^{1+\eta}}{1+\eta} - \rho_2 \frac{h_{2,t}^{1+\gamma}}{1+\gamma} \right]$$

Which simplifies to:

$$U(c_t, h_{1,t}, h_{2,t}) = \frac{(c_t/2)^{1+\eta}}{1+\eta} - \mu \rho_1 \frac{h_{1,t}^{1+\gamma}}{1+\gamma} - (1 - \mu) \rho_2 \frac{h_{2,t}^{1+\gamma}}{1+\gamma}$$

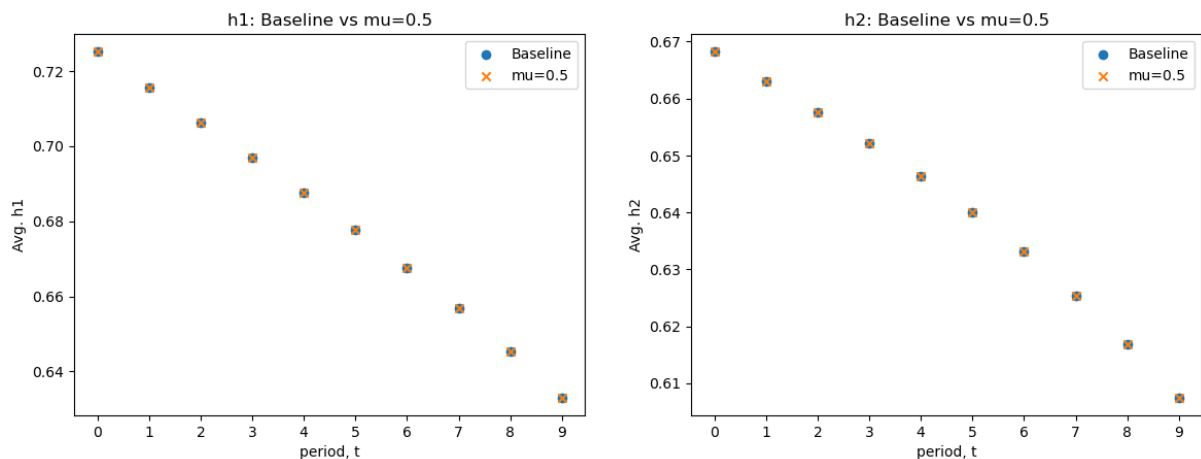


Figure 1: Labor Supply: Baseline vs.  $\mu = 0.5$

2. Why is the behavior unchanged from the modification with  $\mu = 0.5$  in the question above?

The behavior is unchanged as the baseline model from Borella, De Nardi and Yang (2023) already implicitly assumes equal weights of  $\mu = 0.5$  on each individual. The baseline utility function of that model is:

$$U_{baseline} = 2 \frac{(c_t/2)^{1+\eta}}{1+\eta} - \rho_1 \frac{h_{1,t}^{1+\gamma}}{1+\gamma} - \rho_2 \frac{h_{2,t}^{1+\gamma}}{1+\gamma}$$

Recalling the utility function from question 1, we get the following by inserting  $\mu = 0.5$ :

$$U_{modified} = \frac{(c_t/2)^{1+\eta}}{1+\eta} - \frac{1}{2} \rho_1 \frac{h_{1,t}^{1+\gamma}}{1+\gamma} - \frac{1}{2} \rho_2 \frac{h_{2,t}^{1+\gamma}}{1+\gamma}$$

We can see that the two utility functions are proportional as  $\frac{1}{2}U_{baseline} = U_{modified}$ . Since multiplying an objective function by a positive constant does not change the argmax, the optimal hours are identical:

$$\arg \max_{h_{1,t}, h_{2,t}} U_{modified} = \arg \max_{h_{1,t}, h_{2,t}} U_{baseline}$$

3. *Investigate the labor supply responses from individual taxation compared to joint taxation. Concretely, assume that agent 1 starts with human capital of 2 while agent 2 starts with human capital of 0 ( $K_{1,0} = 2$ ,  $K_{2,0} = 0$ ). The tax system under joint taxation is*

$$T^{joint}(Y_1, Y_2) = \left(1 - \lambda^{joint}(Y_1 + Y_2)^{-\tau^{joint}}\right) \cdot (Y_1 + Y_2) \quad (1)$$

*while the individual taxation scheme is*

$$T^{indiv}(Y_1, Y_2) = \left(1 - \lambda^{indiv}(Y_1)^{-\tau^{indiv}}\right) \cdot Y_1 + \left(1 - \lambda^{indiv}(Y_2)^{-\tau^{indiv}}\right) \cdot Y_2 \quad (2)$$

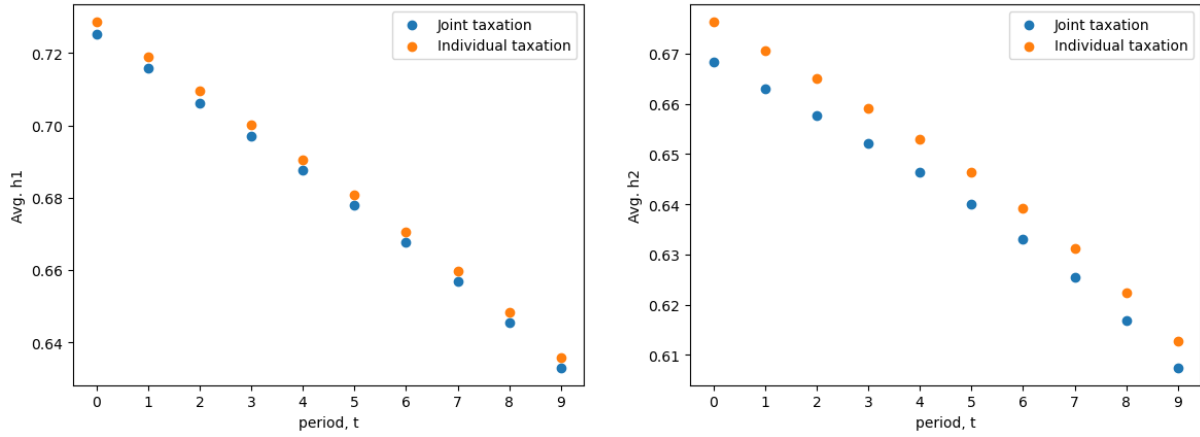
*and (adjusted from Borella, De Nardi and Yang, 2023)*

$$\lambda^{joint} = 2.28, \quad \tau^{joint} = 0.0861765 \quad (3)$$

$$\lambda^{indiv} = 1.75, \quad \tau^{indiv} = 0.0646416 \quad (4)$$

With joint taxation a household is taxed on combined income. This penalizes the household member that earns the least since their income gets added on top of the household member that earns more, pushing them to face a higher marginal tax rate. With individual taxation each household member is taxed individually on their own income, which reduces the marginal tax burden on the member that earns the least, giving them an incentive to work more. In this case, agent 1 starts with more labor market experience, meaning they earn more as they have a higher initial human capital ( $K_{1,0} = 2$ ,  $K_{2,0} = 0$ ).

Figure 2 shows that under individual taxation both agents supply more labor in each period, as each member faces a lower marginal tax rate when taxed on their own income rather than combined household income. The effect is however most pronounced for member 2, who begins with no human capital and therefore has the most to gain from being taxed individually rather than on combined household income.



**Figure 2:** Labor Supply: Joint vs. individual taxation for agent 1 and 2

4. Calculate how the government budget would change from this reform. Be explicit on how you calculate this.

The change in the government budget is simply the difference in total tax revenue collected under joint vs individual taxation. The government budget under each regime is the total taxes collected from the two individuals across all periods.

$$\text{Budget}^{\text{joint}} = \frac{1}{N} \sum_{i=1}^N \sum_{t=0}^T \left( 1 - \lambda^{\text{joint}} (Y_{1,i,t} + Y_{2,i,t})^{-\tau^{\text{joint}}} \right) \cdot (Y_{1,i,t} + Y_{2,i,t})$$

$$\text{Budget}^{\text{indiv}} = \frac{1}{N} \sum_{i=1}^N \sum_{t=0}^T \left[ \left( 1 - \lambda^{\text{indiv}} Y_{1,i,t}^{-\tau^{\text{indiv}}} \right) \cdot Y_{1,i,t} + \left( 1 - \lambda^{\text{indiv}} Y_{2,i,t}^{-\tau^{\text{indiv}}} \right) \cdot Y_{2,i,t} \right]$$

$$\Delta \text{Budget} = \text{Budget}^{\text{indiv}} - \text{Budget}^{\text{joint}}$$

The positive  $\Delta \text{Budget} = 157.76$  from table 1 shows that the government collects more tax revenue under individual taxation despite of each of the members facing a lower marginal tax rate. As we found in Figure 2 both members supply more labor under individual taxation, and the resulting increase in total income more than offsets the lower tax rates.

| Tax Revenue                       |        |
|-----------------------------------|--------|
| Joint taxation                    | 361.05 |
| Individual taxation               | 534.30 |
| Change ( $\Delta \text{Budget}$ ) | 173.25 |

**Table 1:** Government tax revenue under joint and individual taxation

5. Determine at what level of  $\lambda^{\text{indiv}}$  the government budget would not change. Denote this  $\lambda^{\text{indiv},*}$  and be explicit with how you find this value.

We want to find the value of  $\lambda^{\text{indiv}}$  which makes the government budget exactly the same under individual and joint taxation. Since a higher  $\lambda^{\text{indiv}}$  means a lower tax rate, we need to find the value that raises enough tax revenue to match the joint taxation budget. In Question 4 we found that  $\Delta\text{Budget} > 0$ , meaning that individual taxation with  $\lambda^{\text{indiv}} = 1.75$  raised too much tax revenue. We therefore need a higher  $\lambda^{\text{indiv},*}$  to bring revenue down to match joint taxation. This is solved using Brent's method, which results in  $\lambda^{\text{indiv},*} = 1.766990$ , which is very close to the singles tax scale of 1.765038 estimated by Borella, De Nardi and Yang (2023), which makes sense as they should be taxed the same way.

6. Which of the two taxation schemes would a household with initial human capital of  $K_{1,0} = 2$ ,  $K_{2,0} = 0$  (as above) prefer in beginning of life? Use  $\lambda^{\text{indiv},*}$  in  $T^{\text{indiv}}(Y_1, Y_2)$  such that the government budget is the same in the joint and individual taxation scheme. What if both members have high levels of initial human capital, say  $K_{1,0} = 8$ ,  $K_{2,0} = 8$ ?

[Hint: The first-period value function,  $V_0(k_{1,0}, k_{2,0})$ , measures the value of a given set of states (and rules).]

To determine which taxation scheme a household prefers at the beginning of life, we compare the first-period value function  $V_0(k_{1,0}, k_{2,0})$  under each tax system. The household prefers the scheme that yields the higher  $V_0$ , as it represents the higher total discounted lifetime utility.

|                           | $K_{1,0} = 2, K_{2,0} = 0$ | $K_{1,0} = 8, K_{2,0} = 8$ |
|---------------------------|----------------------------|----------------------------|
| $V_0$ joint taxation      | -0.2267                    | -0.1891                    |
| $V_0$ individual taxation | -0.2267                    | -0.1883                    |
| Difference                | 0.00003                    | 0.0007                     |

**Table 2:** First-period value function under joint and individual taxation

In both cases the household prefers individual taxation. However, the preference is stronger when both members have high human capital ( $K_{1,0} = 8, K_{2,0} = 8$ ), with a difference of 0.0007 compared to 0.00003. When both members earn a lot, the penalty from being taxed jointly on combined income is larger, so the gains from being individually taxed are greater.